Global case fatality rate from COVID-19 has decreased by 96.8% during 2.5 years of the pandemic

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We appreciate the commendable work by Wang et al.[1] in evaluating the incidence and case fatality rate (CFR) of SARS-CoV-2 Delta and Omicron variants. Two and a half years have passed since the outbreak of the coronavirus disease 19 (COVID-19) pandemic. The CFR from hospitalized COVID-19 patients was estimated to be 20% or more during the early phase of the pandemic.[2], [3] Over the course of 2.5 years, the mortality rate has drastically decreased. While some believe that COVID-19 is now similar to the common cold or seasonal influenza, others still think of COVID-19 as a highly fatal disease. As the authors stated, the Omicron variant led to a lower CFR around the world. To our knowledge, a chronological study of COVID-19 CFR spanning 2.5 years has not been reported.

The Center for Systems Science and Engineering at Johns Hopkins University, the Center for Systems Science and Engineering provided us with weekly numbers of new cases and deaths from January 26, 2020 to August 21, 2022.[4] Given the time lag between diagnosis and death, CFR was calculated as the "number of deaths in a week" divided by the "number of new cases two weeks prior." A linear regression model was applied after logarithm-transformation of fatality rates. Any week with fewer than 10 deaths or 100 new case was skipped. Besides worldwide data, those for 11 countries with the largest populations were analyzed.

Global case fatalities ranged 1.7%–39.0% in February to March of 2020 and fell below 0.3% in July to August 2022. The coefficient of determinant in the linear regression model was 0.79, and the global CFR was 8.5% in February 2020 and 0.27% in August 2022. The estimated relative risk reduction in 2.5 years was 96.8% (95% confidence interval [CI] 95.6–97.6, $p < 0.001$, Figure 1A).

There was a clear decline in country-level CFR for the 11 included countries with an estimated population size of 4.7 billion (59% of the world's population). More than 90% relative risk reduction during the pandemic was observed in China (97.2%, 95% CI 92.9–98.9, $p < 0.001$, Figure 1B),

USA (90.4%, 95% CI 85.5–93.7, $p < 0.001$, Figure 1D), Brazil (93.6%, 95% CI 90.6–95.6, $p < 0.001$, Figure 1H), Mexico (96.7%, 95% CI 95.1–97.8, $p < 0.001$, Figure 1K), and Japan (98.4%, 95% CI 97.4–99.0, $p < 0.001$, Figure 1L).

We discovered a drastic decline in the global CFR over the last 2.5 years of the COVID-19 pandemic. Fatality rates decreased in all of 11 countries included in this study, despite substantial background social, racial, medical, epidemiological, and methodological inconsistencies among countries. The actual CFR would be even lower since more and more infection cases have gone undiagnosed and unnoticed in more recent months. The decrease in COVID-19 CFR is likely due to multiple factors. Vaccination and postinfection immunity made the disease less severe. Antiviral medications inhibit viral edgrows. Viral lethality mitigated by mutation because host killing is disadvantage for the virus. The potential source of bias was lack of accessibility to test kits at the earlier phase, which may have upshifted the calculated fatality. The COVID-19 CFR is now similar to that of seasonal influenza, which has an estimated CFR in the range of 0.05%–0.5% despite considerable variability depending on the observed year and patient age.[5]

We should recognize that the nature of COVID-19 fatality at the present time greatly differs from that of the early phases of the pandemic.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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